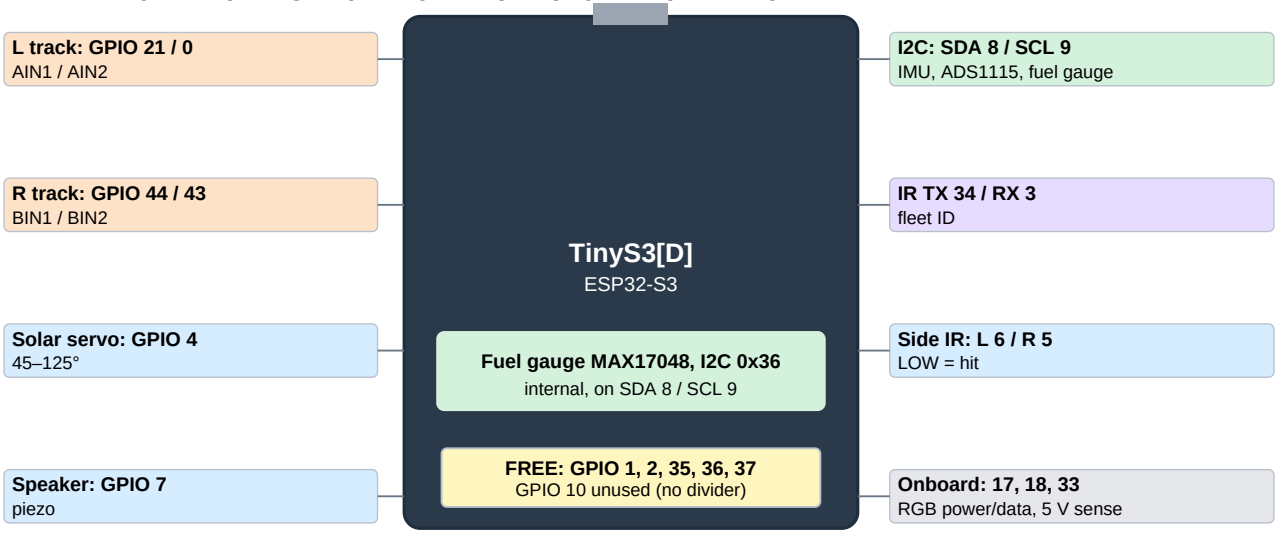
Tracker bot (tracked-bot, /bot2) – pinmap

Unexpected Maker TinyS3[D] (ESP32-S3) | DRV8833 tracks, 130 motors | 1S 800 mAh LiPo + boost converter | firmware env tracked_s3_usb / tracked_s3_ota

1. GPIO table (sorted by function group)

GPIO	Function	Connected part / wire	Notes
0	**Left motor IN2**	DRV8833 AIN2	pos 11. BOOT strap, pulled high at reset (firmware holds it LOW first)
21	**Left motor IN1**	DRV8833 AIN1	pos 10
44	**Right motor IN1**	DRV8833 BIN1	pos 12. UART0 RX. Right track direction inverted in firmware
43	**Right motor IN2**	DRV8833 BIN2	pos 13. UART0 TX, boot log toggles it
4	**Solar servo signal**	Panel tilt servo	pos 8. 45–125°, 90 = flat/centre. Signal released 600 ms after each move
6	**Side IR left**	IR proximity, left	pos 16. Digital, LOW = object (~5 cm), pull-up
5	**Side IR right**	IR proximity, right	pos 9. Digital, LOW = object (~5 cm), pull-up
7	**Speaker**	Passive piezo	pos 17. LEDC PWM
34	**IR TX (fleet ID)**	IR LED + resistor	pos 20. 38 kHz modulated beacon
3	**IR RX (fleet ID)**	TSOP 38 kHz demod	pos 7. LOW = carrier seen. JTAG-source strap, input only
8	**I2C SDA**	IMU, ADS1115, fuel gauge	pos 18. Board default SDA
9	**I2C SCL**	IMU, ADS1115, fuel gauge	pos 19. Board default SCL
17	**RGB LED power**	Onboard	Firmware drives it; off with the LED
18	**RGB LED data**	Onboard NeoPixel	Dim blue idle, green moving, amber OTA
33	**5 V (VBUS) sense**	Onboard	Digital, HIGH = USB/boost 5 V present. Not ADC
10	**Unused**	Nothing connected	No battery divider on this board (onboard fuel gauge instead). Unassigned in firmware
1, 2, 35, 36, 37	**Free**	Nothing connected	Unassigned in firmware

2. Board layout – pins grouped by use (not physical position)



3. I2C bus (SDA GPIO 8, SCL GPIO 9, 3.3 V)

Device	Addr	Notes
MPU-6050 IMU	0x68	AD0 to GND, INT unused
MAX17048 fuel gauge	0x36	Onboard LiPo gauge, same bus. Firmware reads VCELL (0x02) and SOC (0x04) at ~2.5 Hz
ADS1115 ADC	0x48	Auto-probes 0x48–0x4B. Gain ±4.096 V
SSD1306 OLED	0x3C	Optional, probed at boot
BH1750 light	0x23	Optional, probed at boot
VL53L0X ToF	0x29	Optional, probed at boot

4. ADS1115 channels

Ch	Signal	Front end	Ratio
A0	Left BPW34	4.7k to GND, no divider	×1 (pin V)
A1	Right BPW34	4.7k to GND, no divider	×1 (pin V)
A2	Main battery (1S LiPo)	100k top / 100k bottom	×2.0
A3	Solar panel (6 V)	100k from panel+ to pin, 47k pin to GND	×3.128

5. Drive and power

- **DRV8833:** nSLEEP and logic VCC to 3.3 V, VM to boost 5 V, GND common.
- **Track PWM:** 30 kHz, 8-bit, dead-band floor 150/255.
- **Power:** 1S 800 mAh LiPo into the boost converter; 5 V feeds the TinyS3 5 V pin and DRV8833 VM.
- **Battery gauge:** onboard MAX17048 (0x36), read by firmware. Cell volts on `/bot2/battery/voltage`, state of charge % on `/bot2/battery/percent`. Needs a LiPo connected. Battery is also on ADS1115 A2 (`/bot2/power/battery_v`) once wired.
- **Solar panel:** 6 V, 100k from panel+ to ADS A3, 47k from A3 to GND (not wired yet). Topic `/bot2/power/solar_v`.
- **ADS1115 absent:** firmware logs once at boot and publishes no ADS topics.
- **Servo:** boots 45 → 125 → 90, then the signal is released so it draws no idle current.
- **Strap pins:** GPIO 0 (BOOT), 3 (JTAG), 43/44 (UART0) carry boot-time duties; the motor lines are held LOW first in setup().

Source: include/tracked_pins_s3.h, include/tracked_pins.h, src/tracked_main.cpp

Drive | Sensors/actuators | IR comms | I2C | Onboard | Free